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Robust HLL-type Riemann solver capable of resolving contact discontinuity

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ABSTRACT

In this paper, a simple Harten, Lax and van Leer (HLL) type Riemann solver, capable of resolving contact discontinuity accurately, is proposed. Here, the ability to capture the contact is brought about in a way different from Harten, Lax and van Leer with Contact (HLLC) scheme of Toro. In the present method, the flux vector of the Euler equations is split into convective and pressure parts before applying the HLL scheme to each part. Several carefully chosen one- and two-dimensional test problems are solved using the present method in order to demonstrate its robustness and efficacy. The new method is found to be free from problems like spurious expansion shock, carbuncle phenomenon, odd even decoupling, kinked Mach stem, which are usually faced by most of the contact resolving methods.

1. Introduction

Extreme flow conditions are often encountered in high speed flows. Prediction of high speed flows therefore requires accurate, efficient and robust numerical methods. This requirement has led to the development of upwind methods. Over the years, several upwind methods have been proposed, which can be broadly categorized into flux vector splitting (FVS) and flux difference splitting (FDS) methods.

The FVS methods rely on the splitting of flux vectors into upstream and downstream traveling informations. These methods are found to be very successful in capturing steady discontinuities represented by nonlinear waves, which includes shocks. However, they are not effective in capturing the discontinuities represented by linear waves, which result in incorrect diffusion of contact surfaces and shear waves and high dissipation in strong rotational flows [1,2]. For certain applications, such as combustion problems, inappropriate treatment of contact can have undesirable consequences. Since contact surfaces carry discontinuities in temperature and internal energy, ignition criteria can fail due to poor numerical resolution of this kind of wave [3].

On the other hand, the FDS methods are constructed based on approximate solution of local Riemann problem between two adjacent states to calculate the flux through the interface between these states. By virtue of their construction to recognize all wave components in Riemann problem, the FDS methods are capable of capturing discontinuities corresponding to linear as well as nonlinear waves. Due to their superior quality, some of the FDS methods like the Roe's FDS method [4] and Harten, Lax and van Leer with Contact (HLLC) [3] method, have become immensely popular. These methods are capable of capturing sharp and correctly positioned discontinuities of all types such as shock waves and contact surfaces. The Roe's method is based on the Riemann solution of linearized version of Euler equations. This method is capable of capturing any single stationary discontinuity with no numerical dissipation but can generate a nonphysical expansion shock [5]. Such a physically incorrect solution can be avoided by a entropy fix [6]. At strong expansion, however, the Roe method diverges even if the entropy fix is applied [5]. The HLLC method, on the contrary, deals with the original nonlinear Euler equations and possess all desirable properties like positivity and entropy conditions [3]. The HLLC method is considered to be the most preferred FDS method because of its ability to capture discontinuities very accurately. Unfortunately, the HLLC method is found to produce unacceptable results like low frequency post shock fluctuations in case of slowly moving shock, carbuncle phenomenon, odd even decoupling and kinked Mach stem on certain occasions [2,7,8]. In fact, all those methods capable of capturing contact discontinuity accurately referred to as contact-preserving methods [9] face the same difficulty [7]. For example, the approximate FDS methods of Roe [4], Osher and Chakravarthy [10] including exact Riemann solver of Godunov [11] suffer from shock instability and carbuncle phenomenon [7,12–14]. Even the FVS methods when modified to improve its capability to capture contact discontinuity (for example, AUSM [15] method and its family) are not spared from shock instability

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and carbuncle phenomenon [2,7,16]. The modified FVS methods add the ability to capture contact by combining the feature of FDS methods [15].

There have been several attempts in literature to overcome the problem of carbuncle and shock instability. One of the method is based on the observation that the original Harten, Lax and van Leer (HLL) method [17], that was modified by Toro to obtain HLLC scheme [3], do not suffer from shock instability and carbuncle phenomenon. Thus, attempts were made to overcome the problem of anomalous behavior in HLLC method by combining it with HLL scheme [8]. At present, most of the successful attempts [8,12,18] to tackle carbuncle problem are based on such hybridization of complementary solvers for the similar reasons. However, a lot of arbitrariness exist in the choice and design of switching parameters required in the hybrid schemes. For example, the switching over parameter for Roe-HLLE hybrid scheme is problem specific making it complex for practical applications [12]. They may even fail when underlying problem involves interaction of complex flow features [18]. The computational time of hybrid solvers is also higher than that of grid aligned solver [8]. Another successful attempt to tackle carbuncle phenomenon is to use rotated solver. Ren [19] presented rotated Roe Riemann solver and Huang [8] presented rotated HLLC solver to eliminate shock instability. However, this method requires that the numerical flux is computed two times at each grid face, which makes the rotated methods less efficient computationally. Furthermore, the rotated solvers are tested for two-dimensional problems only. Although partially successfull in avoiding carbuncle, the AUSMDV, HUS (AUSM – Roe/Osher hybrid) and AUSM$^+$ schemes exhibit the carbuncle phenomenon under some conditions [2,20]. Very recently, Sun et al. is able to successfully take care of these instabilities with greater success by using artificial wave formulation, called AUFS [13]. However, the idea behind the choice of these artificial waves is not very clear. Also, the AUFS is found to introduce extra numerical diffusion. There exists not a single accurate contact preserving method which is completely free from the problem of shock instability and carbuncle phenomenon. It is therefore necessary to look for an alternative approach.

In this paper, a very simple novel methodology based on HLL-type Riemann solver is proposed which does not suffer from numerical shock instability problem. Realizing that the problem of the anomalous behavior is manifested only when the contact wave is restored into the HLLC method following the same principles as in the original HLL solver, a different approach is attempted in the present work in order to introduce the contact surface. Here, the contact surface is restored by using convection pressure split form of Euler equations following the approaches of Zha-Bilgen [21], AUSM [15] and CUSP [22] type of schemes. The new method henceforth will be referred to as HLL convective-pressure split (HLL-CPS) scheme. In HLL-CPS, the convective part is discretized in a (FDS) manner similar to Zha-Bilgen, AUSM and CUSP type of schemes; that is, the complete convective part is evaluated based on left state or right state depending on the sign of local fluid velocity. The pressure part in HLL-CPS is discretized using HLL-type two wave approximation. Since, the proposed scheme has been developed as an alternative to HLLC scheme that can resolve contact accurately, the accuracy and efficiency of the proposed scheme have been compared with the HLLC scheme.

The paper is organized in the following manner: After a brief description of convective pressure splitting strategies of one-dimensional (1D) Euler equations in Section 2, finite volume discretization of 1D Euler equations using HLL-CPS schemes is described in detail in Section 3. Extension of the HLL-CPS schemes to two-dimensional (2D) case is explained in Section 4. Numerical results for several 1D and 2D test cases are presented in Section 5 followed by conclusion in Section 6. The numerical results of the present method are compared with HLL, HLLC and AUFS schemes to show its relative merits and demerits.

2. Governing equations

In the present approach, Convective Pressure Split (CPS) Euler equations of the form:

$$\frac{\partial \mathbf{U}}{\partial t} + \frac{\partial \mathbf{F}_1}{\partial x} + \frac{\partial \mathbf{F}_2}{\partial x} = 0 \tag{1}$$

are considered, where $\mathbf{U} = [\rho \quad \rho u \quad \rho e]^T$ is the vector of conserved quantities, $\mathbf{F}_1$ and $\mathbf{F}_2$ are the split flux vectors. The convective-pressure splitting can be done in two ways; one following Zha-Bilgen [21] formulation (referred to as CPS-Z) and the other following AUSM [15] formulation (referred to as CPS-A). The split flux vectors in case of CPS-Z are given as:

$$\mathbf{F}_1 = u\begin{pmatrix} \rho \\ \rho u \\ \rho e \end{pmatrix} \quad \mathbf{F}_2 = \begin{pmatrix} 0 \\ p \\ pu \end{pmatrix} \tag{2}$$

whereas, that in CPS-A are given as

$$\mathbf{F}_1 = u\begin{pmatrix} \rho \\ \rho u \\ \rho e + p \end{pmatrix} \quad \mathbf{F}_2 = \begin{pmatrix} 0 \\ p \\ 0 \end{pmatrix} \tag{3}$$

The associated wavespeeds corresponding to the above convective flux component $\mathbf{F}_1$ are (u, u, u) and those of pressure flux component $\mathbf{F}_2$ are considered as ($-a$, 0, a) for both type of splittings [15,21,22]. This implies that the information of the convective terms propagates uniformly in the same direction as the velocity vector u goes, and the information of the pressure terms travels with the convective terms at the speed u and propagates in all directions at the sound speed a [21].

3. Finite volume discretisation of 1D Euler equations

Considering the integral form of the 1D Euler Eq. (1)

$$\frac{\partial}{\partial t}\int_{\Omega} \mathbf{U}\, d\Omega + \oint_{\partial\Omega} (\mathbf{F}_1 + \mathbf{F}_2)\cdot \mathbf{n}\, ds = 0 \tag{4}$$

finite volume discretisation can be written as

$$\frac{\partial \bar{\bar{\mathbf{U}}}}{\partial t}\Omega_i + \sum_{l=1}^{2} (\mathbf{F}_1 + \mathbf{F}_2)_l \cdot \mathbf{n}_l \Delta s_l = 0 \tag{5}$$

where quantity with superscript double bar indicates cell average value, quantities with subscript l indicate values associated with the face l enclosing the finite volume Ω_i, Δs_l is unity and Ω_i is Δx in case of 1D. Discretizing time derivative, above equation can be written as

$$\bar{\bar{\mathbf{U}}}_i^{n+1} = \bar{\bar{\mathbf{U}}}_i^{n} - \frac{\Delta t}{\Delta x}\sum_{l=1}^{2} (\mathbf{F}_1 + \mathbf{F}_2)_l \cdot \mathbf{n}_l \Delta s_l = 0 \tag{6}$$

where superscript n indicates time level and Δt is time step. The cell face normal vector $\mathbf{n}_l$ is taken 1 for right face and -1 for left face. Fluxes at interface l are evaluated using new **HLL**-type Riemann solvers for the **C**onvective **P**ressure **S**plit Euler equations called HLL-CPS schemes.

3.1. HLL-CPS schemes

Two HLL-CPS schemes are described here; one using the CPS-Z version of flux splitting (2) and the other using the CPS-A version of flux splitting (3). Since the total flux vector (in both the schemes) has been split into two parts, the numerical treatment for each part has been described separately.

3.1.1. Evaluation of flux vector F_1 at cell interface

The flux vector $\mathbf{F}_1$ for CPS-Z splitting (2) can be rewritten as:

$$\mathbf{F}_1 = M\begin{pmatrix} \rho a \\ \rho u a \\ \rho e a \end{pmatrix} \tag{7}$$

where a is the speed of sound and M is the local Mach number. The eigenvalues of $\mathbf{F}_1$ (i.e. u, u, u) are either non-negative or non-positive. Thus the direction of propagation of convective flux component is entirely dependent on the Mach number M only. The interface convective flux can be calculated as

$$\mathbf{F}_{1\left(\frac{1}{2}\right)} = M_k\begin{pmatrix} \rho \\ \rho u \\ \rho e \end{pmatrix}_k a_k \tag{8}$$

$$k = \begin{cases} L & \text{if } \bar{u} \geqslant 0 \\ R & \text{if } \bar{u} < 0 \end{cases} \tag{9}$$

where $\bar{u} = \frac{u_L+u_R}{2}$

$$M_k = \begin{cases} \frac{\bar{u}}{\bar{u}-S_L} & \text{if } \bar{u} \geqslant 0 \\ \frac{\bar{u}}{\bar{u}-S_R} & \text{if } \bar{u} < 0 \end{cases} \tag{10}$$

and

$$a_k = \begin{cases} u_L - S_L & \text{if } \bar{u} \geqslant 0 \\ u_R - S_R & \text{if } \bar{u} < 0 \end{cases} \tag{11}$$

Above S_L and S_R are the carefully selected wavespeeds which are discussed later in Section 3.2.

The interface convective flux $\mathbf{F}_1$ for CPS-A splitting (3) can be similarly evaluated as:

$$\mathbf{F}_{1\left(\frac{1}{2}\right)} = M_k\begin{pmatrix} \rho \\ \rho u \\ \rho e + p \end{pmatrix}_k a_k \tag{12}$$

where left and right state are evaluated based on the condition given in (9). The values of M_k and a_k are specified by the expressions (10) and (11) respectively.

3.1.2. Evaluation of flux vector F_2 at cell interface

The flux vector $\mathbf{F}_2$ has been evaluated based on HLL-type Riemann solver [17]. The HLL flux vector for $\mathbf{F}_2$ at the interface is given by:

$$\mathbf{F}_{2\left(\frac{1}{2}\right)} = \frac{S_R\mathbf{F}_{2L} - S_L\mathbf{F}_{2R} + S_RS_L(\mathbf{U}_R - \mathbf{U}_L)}{S_R - S_L} \tag{13}$$

which can also be written as

$$\mathbf{F}_{2\left(\frac{1}{2}\right)} = \frac{1}{2}(\mathbf{F}_{2L} + \mathbf{F}_{2R}) + \delta\mathbf{U}_2 \tag{14}$$

where $\delta\mathbf{U}_2$ is the numerical diffusion due to $\mathbf{F}_2$. The fastest left and right running wave speeds S_L and S_R are evaluated keeping in mind that the information of the pressure terms goes with the convective terms at the speed u and propagates in all directions at the sound speed a [21]. The estimations of waves speeds are given in Section 3.2. Now, from the Eq. (13), the expression for $\delta\mathbf{U}_2$ can be derived as:

$$\delta\mathbf{U}_2 = \frac{S_R + S_L}{2(S_R - S_L)}(\mathbf{F}_{2L} - \mathbf{F}_{2R}) - \frac{S_LS_R}{S_R - S_L}\begin{pmatrix} \rho_L - \rho_R \\ (\rho u)_L - (\rho u)_R \\ (\rho e)_L - (\rho e)_R \end{pmatrix} \tag{15}$$

It may be noted at this stage, the HLL schemes for CPS-Z and CPS-A splitting are respectively named as HLL-CPS-Z and HLL-CPS-A schemes.

Since density changes are involved across any contact discontinuity, it is clear from above expression (15) that numerical diffusion will be introduced across a contact discontinuity by the above expression of the HLL scheme. In order to avoid the numerical diffusion and capture the contact discontinuity accurately, isentropic conditions are assumed and density terms are replaced with pressure terms in a manner similar to reference [13]. This assumption is true for most of the practical problems. In case of isentropic condition imposed by the expression $\bar{a}^2 = \delta p/\delta\rho$ [13], above numerical diffusion (15) gets modified as

$$\delta\mathbf{U}_2 = \frac{S_R + S_L}{2(S_R - S_L)}(\mathbf{F}_{2L} - \mathbf{F}_{2R}) - \frac{S_LS_R}{\bar{a}^2(S_R - S_L)} \times \begin{pmatrix} p_L - p_R \\ (pu)_L - (pu)_R \\ \frac{\bar{a}^2}{\gamma-1}(p_L - p_R) + \frac{1}{2}\left[(pu^2)_L - (pu^2)_R\right] \end{pmatrix} \tag{16}$$

where $\bar{a} = \frac{a_L+a_R}{2}$

3.1.3. Evaluation of total flux vector at cell interface

The total flux vector at a cell interface can thus be written as

$$\mathbf{F}_{\left(\frac{1}{2}\right)} = \mathbf{F}_{1\left(\frac{1}{2}\right)} + \frac{1}{2}(\mathbf{F}_{2L} + \mathbf{F}_{2R}) + \delta\mathbf{U}_2 \tag{17}$$

1. In case of HLL-CPS-Z scheme, $\mathbf{F}_{1\left(\frac{1}{2}\right)}$ is evaluated using (8), $\mathbf{F}_{2k}(k = L$ or $R)$ using (2) and $\delta\mathbf{U}_2$ using (16).
2. In case of HLL-CPS-A scheme, $\mathbf{F}_{1\left(\frac{1}{2}\right)}$ is evaluated using (12), $\mathbf{F}_{2k}(k = L$ or $R)$ using (3) and $\delta\mathbf{U}_2$ using (16).

3.2. Selection of wavespeeds

Wavespeed selection can be done in two ways:

1. First method is based on Roe average variables, i.e.

$$S_L = \min(0, u_L - a_L, \tilde{u} - \tilde{a}), \quad S_R = \max(0, u_R + a_R, \tilde{u} + \tilde{a}) \tag{18}$$

where $\tilde{u}$ = Roe averaged velocity and $\tilde{a}$ = Roe averaged speed of sound. Zero has been added in the above expressions (18) in order to ensure fully one sided flux in supersonic regime. The HLL-CPS schemes using this type of wavespeed selection have been referred as HLL-CPS-Z_{roe} and HLL-CPS-A_{roe}.

2. Another method of selection of wavespeeds is based on isentropic calculations, i.e.

$$S_L = \min(0, u_L - a_L, u^\star - a^\star), \quad S_R = \max(0, u_R + a_R, u^\star + a^\star) \tag{19}$$

where $u^\star = \frac{u_L+u_R}{2} + \frac{a_L-a_R}{\gamma-1}$ and $a^\star = \frac{a_L+a_R}{2} + \frac{\gamma-1}{4}(u_L - u_R)$.
The HLL-CPS schemes using this type of wavespeed selection have been referred as HLL-CPS-Z_{isen} and HLL-CPS-A_{isen}.

For both the type of wavespeed selections, if $\bar{u}$ is zero, then wavespeeds are modified as

$$S_L = -\bar{a} \quad \text{and} \quad S_R = \bar{a} \tag{20}$$

A sample algorithm for HLL-CPS-Z_{roe} based on 17,18,20 is described in Appendix A.

Table 1
Data for 1D shock tube test cases.

Test	$(\rho, u, p)_L$	$(\rho, u, p)_R$	x_0	t
1	(1.0, 0.75, 1.0)	(0.125, 0.0, 0.1)	0.3	0.2
2	(1.0, −2.0, 0.4)	(1.0, 2.0, 0.4)	0.5	0.15
3	(1.0, 0.0, 1000.0)	(1.0, 0.0, 0.01)	0.5	0.012
4	(5.99924, 19.5975, 460.894)	(5.99242, −6.19633, 46.095)	0.4	0.035
5	(1.0, −19.59745, 1000.0)	(1.0, −19.59745, 0.01)	0.8	0.012
6	(1.4, 0.0, 1.0)	(1.0, 0.0, 1.0)	0.5	2
7	(1.4, 0.1, 1.0)	(1.0, 0.1, 1.0)	0.5	2
8	(1.0, 2.0, 0.1)	(1.0, −2.0, 0.1)	0.5	0.8

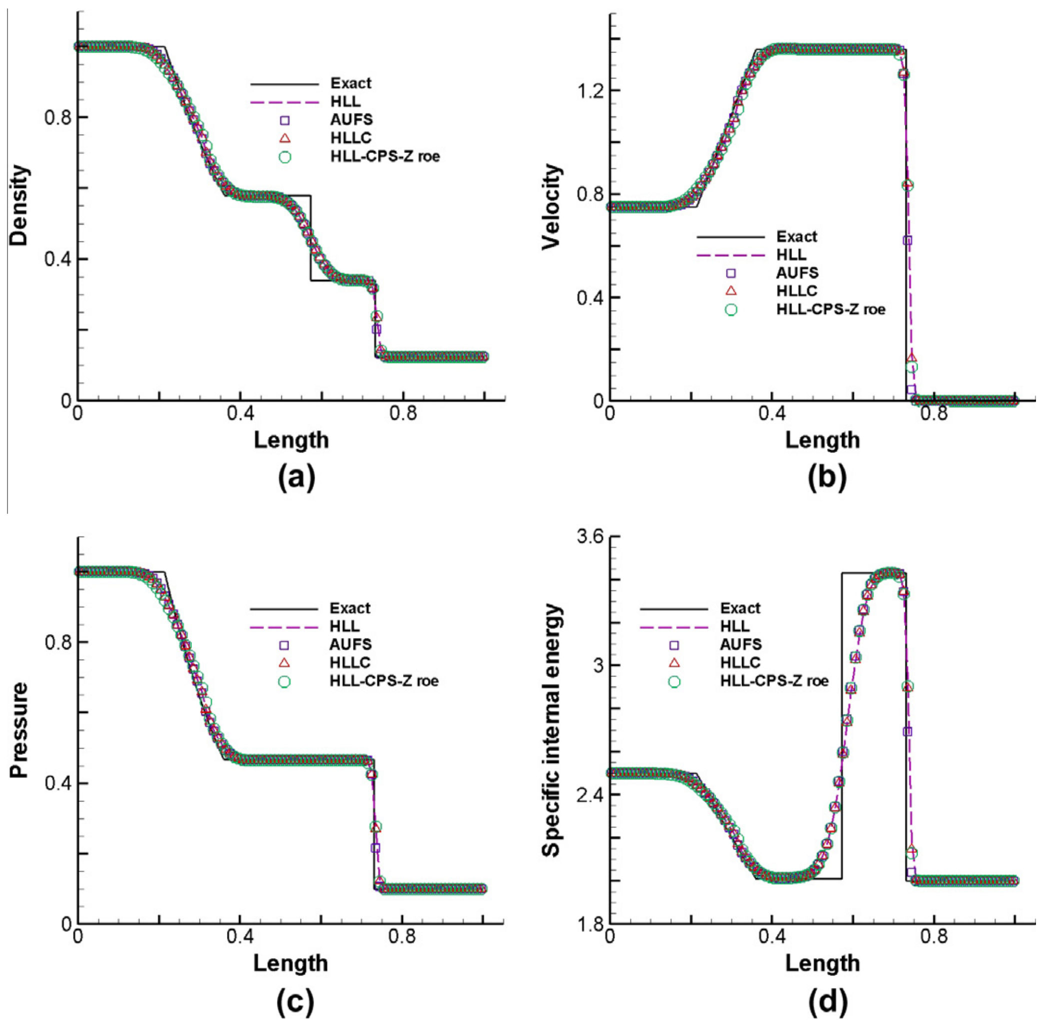


Fig. 1. Test case 1.

4. Extension of HLL-CPS schemes to 2D finite volume method

Consider integral form of 2D Euler equations

$$\frac{\partial}{\partial t}\int_{\Omega} \mathbf{U} d\Omega + \oint_{\partial\Omega} (\mathbf{F} + \mathbf{G}) \cdot \mathbf{n} \, ds = 0 \tag{21}$$

where $\mathbf{n} = (n_x, n_y)$ is the face (edge in 2D) normal, $\mathbf{U} = [\rho \quad \rho u \quad \rho v \quad \rho e]^T$ is the vector of conserved quantities. Face normal flux vector $\mathcal{F}$ is split into convective part $\mathcal{F}_1$ and pressure part $\mathcal{F}_2$ as done in 1D formulation, i.e.

$$\mathcal{F} \equiv (\mathbf{F} + \mathbf{G}) \cdot \mathbf{n} = \mathcal{F}_1 + \mathcal{F}_2$$

where split normal fluxes in case of CPS-Z are given by:

$$\mathcal{F}_1 = u_n \begin{pmatrix} \rho \\ \rho u_n \\ \rho u_t \\ \rho e \end{pmatrix} \qquad \mathcal{F}_2 = \begin{pmatrix} 0 \\ p \\ 0 \\ p u_n \end{pmatrix} \tag{22}$$

whereas split normal fluxes in case of CPS-A are given by:

$$\mathcal{F}_1 = u_n \begin{pmatrix} \rho \\ \rho u_n \\ \rho u_t \\ \rho e + p \end{pmatrix} \qquad \mathcal{F}_2 = \begin{pmatrix} 0 \\ p \\ 0 \\ 0 \end{pmatrix} \tag{23}$$

where normal velocity $u_n = u n_x + v n_y$ and tangential velocity $u_t = -u n_y + v n_x$. For both CPS-Z and CPS-A splittings the wavespeeds

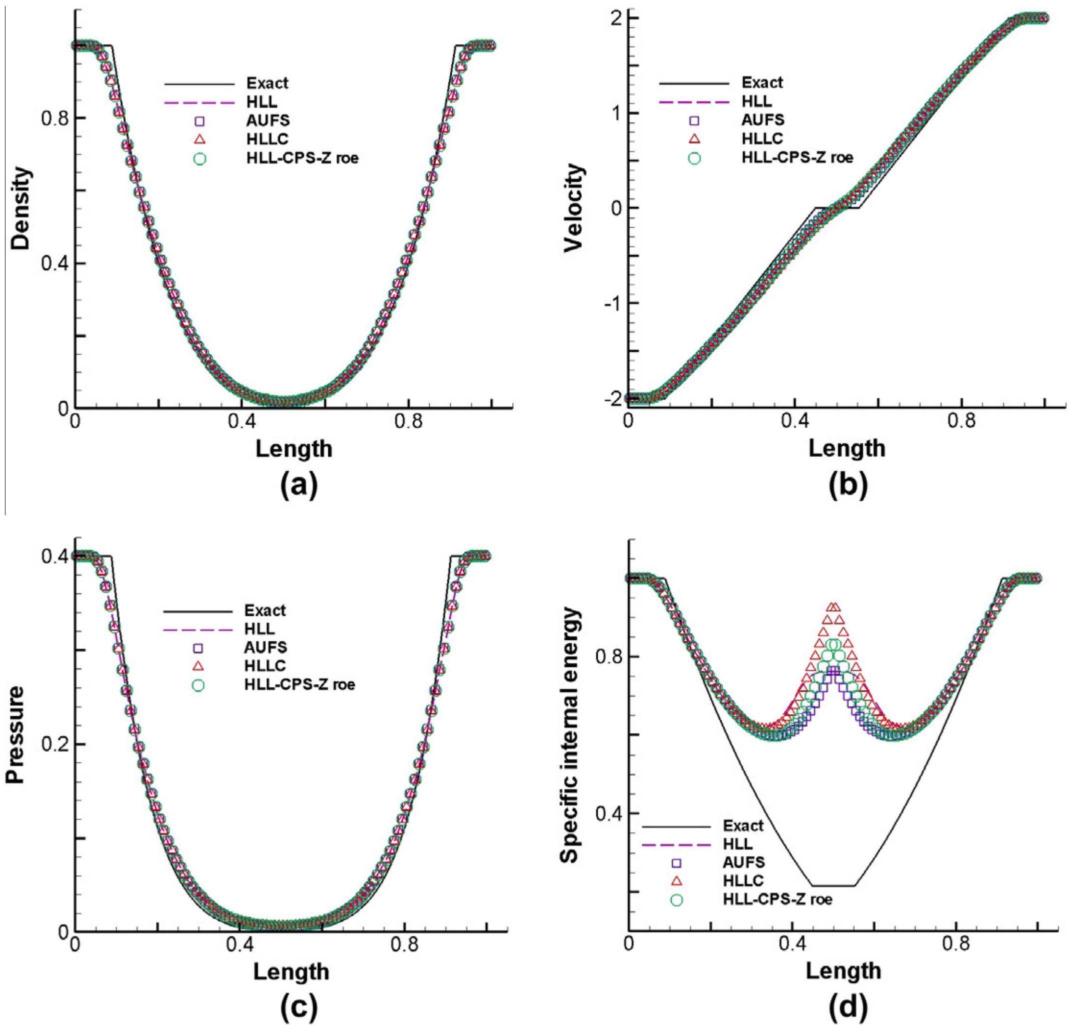


Fig. 2. Test case 2.

corresponding to $\mathcal{F}_1$ are (u_n, u_n, u_n, u_n), and those in case of $\mathcal{F}_2$ are ($-a$, 0, 0, a).

The finite volume discretization of (21) can now be written as:

$$\overline{\overline{\mathbf{U}}}_{i,j}^{n+1} = \overline{\overline{\mathbf{U}}}_{i,j}^{n} - \frac{\Delta t}{\Omega_{i,j}} \sum_{l=1}^{nf} (\mathcal{F}_1 + \mathcal{F}_2)_l \Delta s_l = 0 \tag{24}$$

Δs_l is the edge length, nf is the number of edges enclosing the 2D finite volume $\Omega_{i,j}$. The numerical flux $\mathcal{F}$ is obtained by solving a Riemann problem in the direction normal to the interface and then rotating it back to get grid aligned flux. Based on the analysis similar to 1D formulations, the total numerical flux can be written as

$$\mathcal{F}_l = \mathcal{F}_{1\left(\frac{1}{2}\right)} + \frac{1}{2}[\mathcal{F}_{2L} + \mathcal{F}_{2R}] + \delta\mathcal{U}_2 \tag{25}$$

where interface convective flux $\mathcal{F}_{1\left(\frac{1}{2}\right)}$, the numerical diffusion $\delta\mathcal{U}_2$ are described below. In the above expression (25), $\mathcal{F}_{2k}(k = L \text{ or } R)$ is evaluated using (22) or (23) for HLL-CPS-Z or HLL-CPS-A formulations respectively.

4.1. Evaluation of interface convective flux $\mathcal{F}_{1\left(\frac{1}{2}\right)}$

1. In case of HLL-CPS-Z, interface convective flux can be evaluated as

$$\mathcal{F}_{1\left(\frac{1}{2}\right)} = M_{nk} \begin{pmatrix} \rho \\ \rho u_n \\ \rho u_t \\ \rho e \end{pmatrix}_k a_k \tag{26}$$

$$k = \begin{cases} L & \text{if } \bar{u}_n \geqslant 0 \\ R & \text{if } \bar{u}_n < 0 \end{cases} \tag{27}$$

where $\bar{u}_n = \frac{u_{nL}+u_{nR}}{2}$

$$M_{nk} = \begin{cases} \frac{\bar{u}_n}{\bar{u}_n - S_L} & \text{if } \bar{u}_n \geqslant 0 \\ \frac{\bar{u}_n}{\bar{u}_n - S_R} & \text{if } \bar{u}_n < 0 \end{cases} \tag{28}$$

and

$$a_k = \begin{cases} u_{nL} - S_L & \text{if } \bar{u}_n \geqslant 0 \\ u_{nR} - S_R & \text{if } \bar{u}_n < 0 \end{cases} \tag{29}$$

The S_L and S_R are the selected wavespeeds which are discussed later in Section 4.3.

2. In case of HLL-CPS-A, interface convective flux can be evaluated as

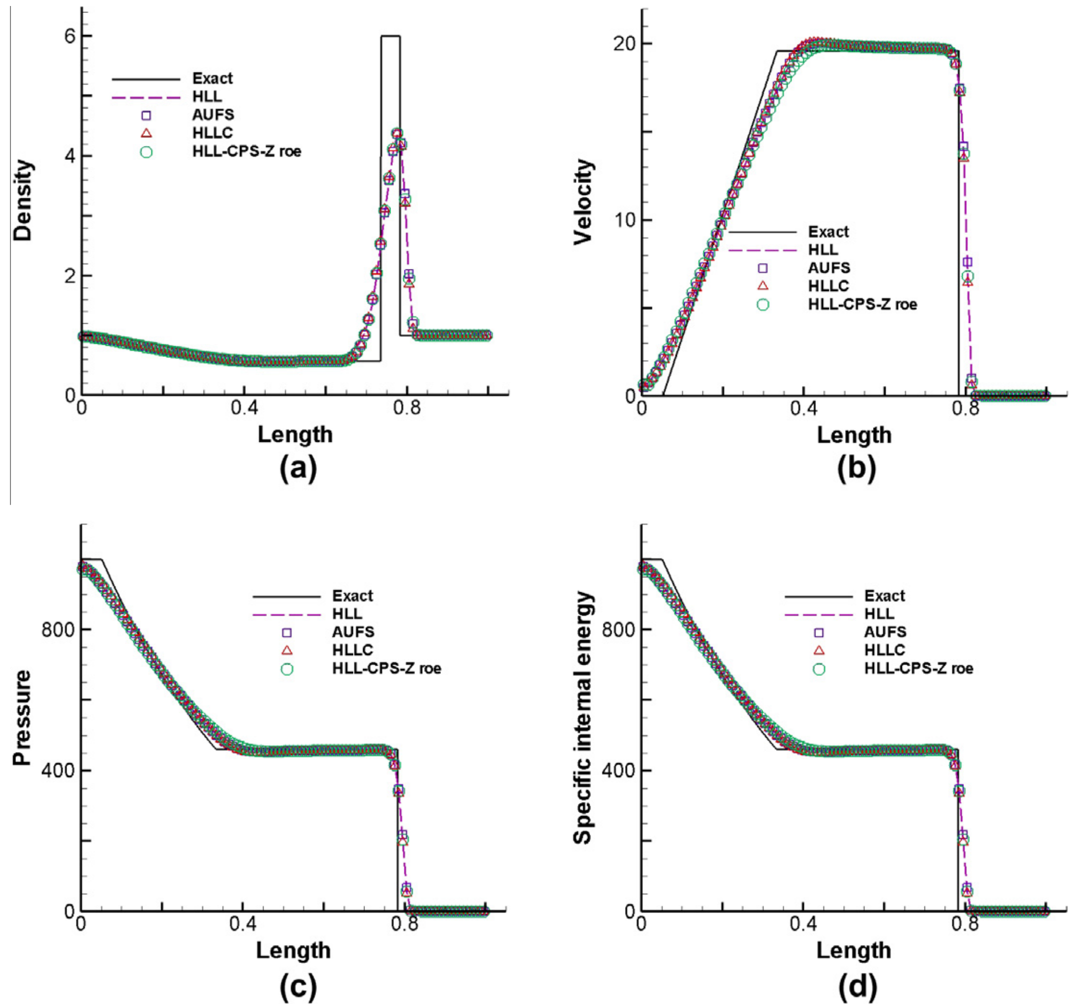


Fig. 3. Test case 3.

$$\mathcal{F}_{1(\frac{1}{2})} = M_{nk} \begin{pmatrix} \rho \\ \rho u_n \\ \rho u_t \\ \rho e + p \end{pmatrix}_k a_k \tag{30}$$

The left and right state are evaluated based on the condition given in (27). The values of M_{nk} and a_k are specified by the expressions (28) and (29), respectively.

4.2. Evaluation of numerical diffusion $\delta\mathcal{U}_2$

The numerical diffusion is calculated as given below:

$$\delta\mathcal{U}_2 = \frac{S_R + S_L}{2(S_R - S_L)}(\mathcal{F}_{2L} - \mathcal{F}_{2R}) - \frac{S_L S_R}{\bar{a}^2(S_R - S_L)} \times \begin{pmatrix} p_L - p_R \\ (pu_n)_L - (pu_n)_R \\ (pu_t)_L - (pu_t)_R \\ \frac{\bar{a}^2}{\gamma-1}(p_L - p_R) + \frac{1}{2}[(pq^2)_L - (pq^2)_R] \end{pmatrix} \tag{31}$$

where $q^2 = u_n{}^2 + u_t{}^2$. In the above expression (31), $\mathcal{F}_{2k}(k = L \text{ or } R)$ is evaluated using (22) or (23) for HLL-CPS-Z or HLL-CPS-A formulations respectively. The estimation of wave speeds S_L and S_R is described below.

4.3. Methods of wave speed estimation

- The first method is based on Roe average variables, i.e.

$$S_L = \min(0, u_{nL} - a_L, \tilde{u}_n - \tilde{a}), \quad S_R = \max(0, u_{nR} + a_R, \tilde{u}_n + \tilde{a}) \tag{32}$$

where $\tilde{u}_n$ is the Roe average velocity based on u_{nL} and $u_{nR}, \tilde{a}$ is Roe averaged speed of sound.

- The second method is based on isentropic calculations, i.e.

$$S_L = \min\left(0, u_{nL} - a_L, u_n^\star - a^\star\right), \quad S_R = \max\left(0, u_{nR} + a_R, u_n^\star + a^\star\right) \tag{33}$$

where $u_n^\star = \frac{u_{nL}+u_{nR}}{2} + \frac{a_L - a_R}{\gamma-1}, a^\star = \frac{a_L+a_R}{2} + \frac{\gamma-1}{4}(u_{nL} - u_{nR})$.

- For both the above methods, if $\bar{u}_n$ is zero, then wavespeeds are modified as

$$S_L = -\bar{a} \quad \text{and} \quad S_R = \bar{a} \tag{34}$$

5. Numerical results

The performances of the HLL-CPS-Z and HLL-CPS-A schemes have been evaluated using several carefully chosen 1D and 2D test cases for Euler equations (with $\gamma = 1.4$) using both methods of wavespeed selection as described in Sections 3.2 and 4.3. The test problems chosen here are reported to produce computational difficulties for most of the advanced upwind methods. The

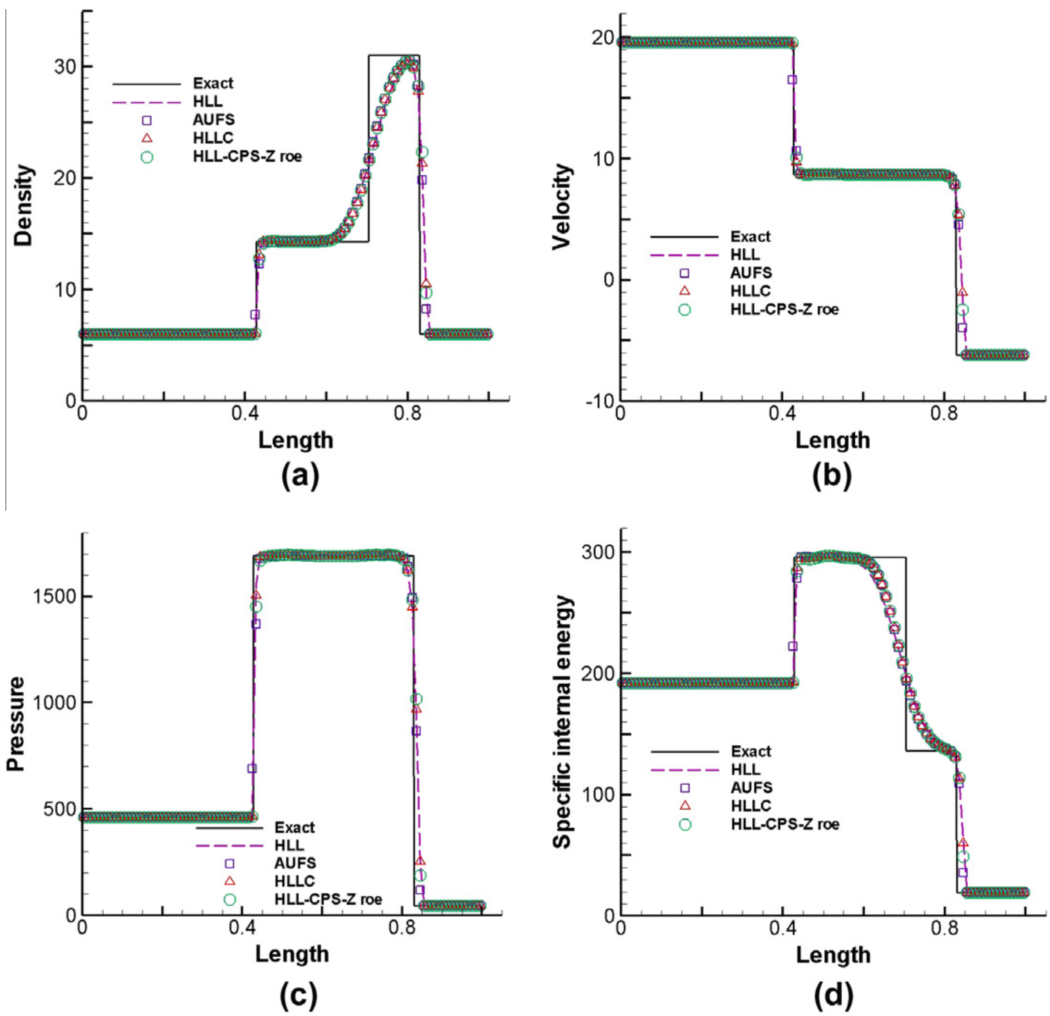


Fig. 4. Test case 4.

HLL-CPS-Z scheme is found to be marginally more accurate than the HLL-CPS-A scheme. Among the HLL-CPS-Z_{roe} and HLL-CPS-Z_{isen} schemes, the first one being more accurate, results of HLL-CPS-Z_{roe} scheme only are presented for all the test cases. Only results of HLL-CPS-Z_{isen} scheme for slowly moving shock and blunt body problems are reported here as these results show little superiority over the other. The results of the present method is always compared with original HLL, HLLC methods to show relative improvement in its solutions. The left running wavespeed S_L and right running wavespeed S_R for HLL and HLLC have been chosen as

$$S_L = \min(u_L - a_L, \tilde{u} - \tilde{a}), \quad S_R = \max(u_R + a_R, \tilde{u} + \tilde{a})$$

where superscript *tilde* indicates Roe averaged quantity. The wavespeed of middle wave S_M for HLLC has been taken as

$$S_M = \frac{p_R - p_L + \rho_L u_L (S_L - u_L) - \rho_R u_R (S_R - u_R)}{\rho_L (S_L - u_L) - \rho_R (S_R - u_R)}$$

The HLL-CPS results are also compared with the most robust and accurate solver AUFS and the exact solutions whenever available.

As discretization errors among first order schemes are easily distinguishable, we have presented only the first order results here. In almost all the test cases, the second order computations for all the schemes are found to improve in terms of resolution as expected.

5.1. One dimensional test cases

A series of shock tube problems and a slowly moving shock problem have been presented here.

5.1.1. Shock tube problems

Eight test cases [23] for shock tube problem have been considered for rigorous testing of the numerical schemes. Initial condition for a shock tube problem consists of two states, i.e. left state and right state separated by a diaphragm (discontinuity). The values of left state denoted by $(\rho, u, p)_L$, right state by $(\rho, u, p)_R$ and the location of the discontinuity denoted by x_0 for eight test cases are listed in Table 1. Since, the waves do not reach the left and right boundaries at the specified time t, the left and right boundary values are extrapolated from the adjacent cell of the domain. Being unsteady, the results have been presented for time t as given in the same Table 1. For all the test cases, exact solution and numerical solutions of HLL, HLLC, AUFS and HLL-CPS-Z_{roe} schemes have been presented. The computational domain of length unity consists of 100 cells and Courant–Friedrichs–Lewy (CFL) number of 0.9 has been used. The exact solution has been presented for 2000 cells to get crisp discontinuities.

Test case 1 has a right running shock, a right running contact discontinuity and a left running expansion wave. The expansion region has a sonic point inside. As seen in Fig. 1, the results of HLL, HLLC, AUFS and HLL-CPS-Z_{roe} schemes are coinciding at shock, contact

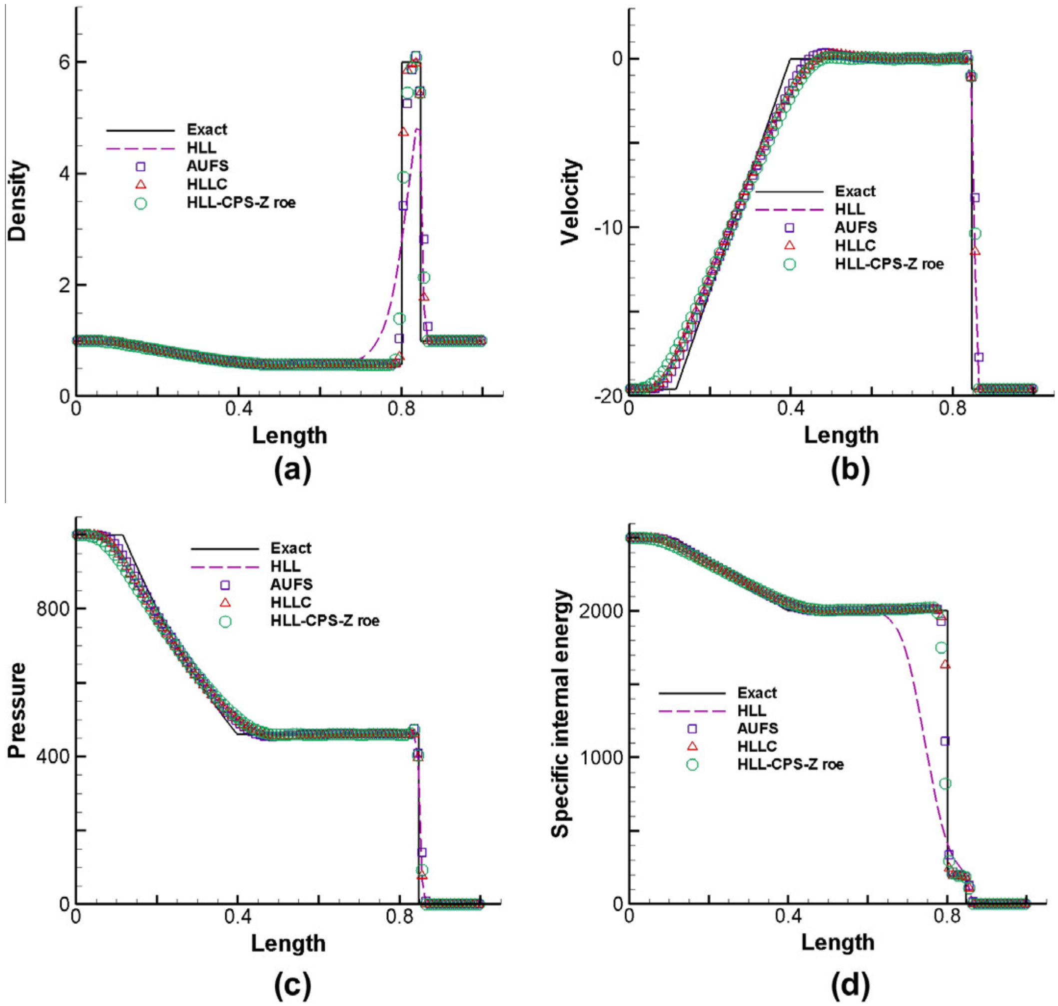


Fig. 5. Test case 5.

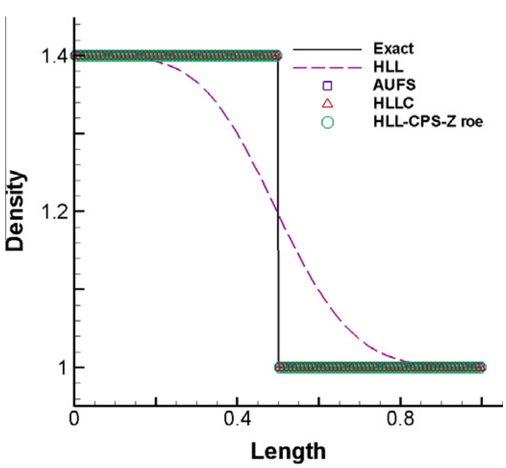


Fig. 6. Test case 6.

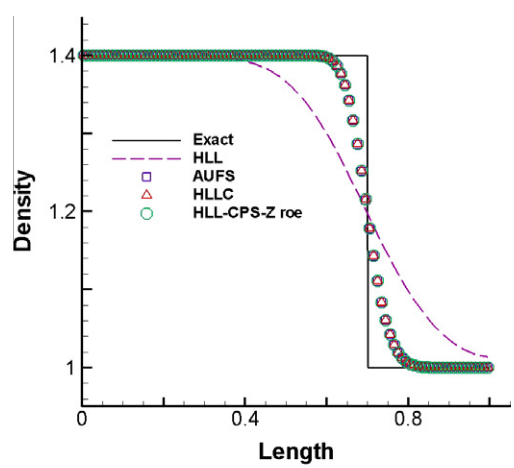


Fig. 7. Test case 7.

and expansion fan. All the schemes take four cells to resolve shock and 20 cells to resolve contact. HLL-CPS-Z_{roe} scheme is able to resolve the sonic point smoothly and accurately; thus, it is able to overcome the entropy glitch [23]. The solution of HLL-CPS-Z_{roe} scheme is found to be monotone without any spurious oscillations.

Test case 2 is a receding flow problem. It has two expansion waves running outwards either side and a stationary contact. In this problem, both density and pressure are close to zero around contact and a small error in these variables is over magnified in the internal energy plot [23]. As seen in Fig. 2, the solutions for

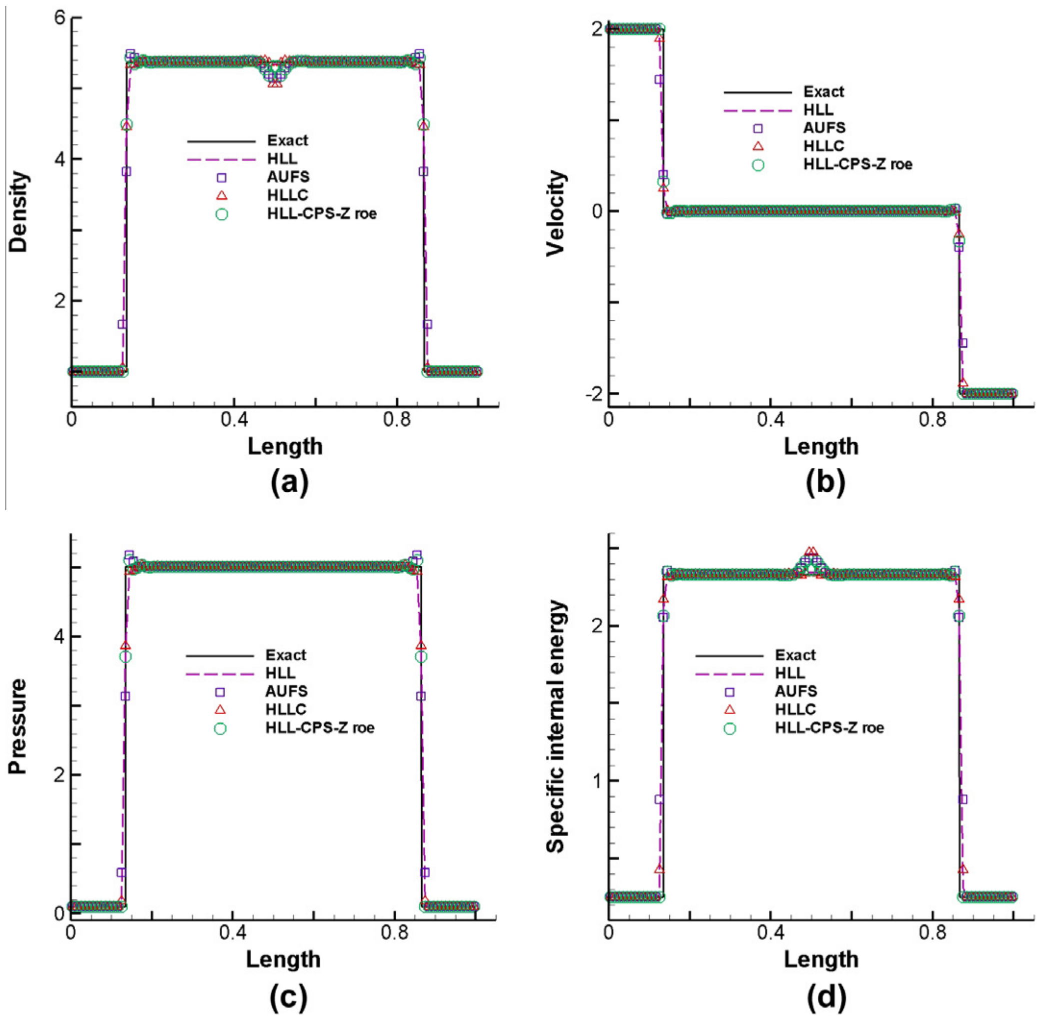


Fig. 8. Test case 8.

pressure, velocity and density are almost coinciding for all the schemes. The internal energy is best resolved by AUFS scheme. If internal energy plot is analyzed by integrating the area under the curve, then HLL-CPS-Z_{roe} solution is slightly inferior to that of AUFS but it is better than HLL and HLLC. However, AUFS scheme is able to retain this edge in first order result only. In second order result, internal energy solution using all four schemes produces similar area under the curve.

Test case 3 has a Mach 198 strong shock running rightwards, a right running contact and a left running expansion wave. As seen in Fig. 3, the resolution of expansion fan obtained from HLL, HLLC and AUFS schemes are almost similar, whereas result of HLL-CPS-Z_{roe}scheme show little extra diffusion at tail portion of the expansion fan. However, HLL, HLLC and AUFS results show little overshoot at the tail of expansion fan; whereas HLL-CPS-Z_{roe} result is monotone and smoothly merging with exact solution. The HLL scheme lacks in resolving the contact. Shock wave has been properly resolved by all the schemes.

Test case 4 has two shocks and one contact. All three discontinuities are moving towards right, but the left shock is slowly moving. Quirk had reported low [12] frequency noise with some solvers while solving slowly moving shocks. As seen in Fig. 4 slowly moving left shock has been resolved by HLL-CPS-Z_{roe} scheme without any oscillations. The HLL, HLLC and HLL-CPS-Z_{roe} schemes are able to resolve slow moving shock over three cells, whereas AUFS scheme takes four cells. The AUFS scheme has extra diffusion at slow moving shock. The contact is diffusive with HLL scheme. Right shock has been resolved similarly with all the four schemes.

Test case 5 has a strong shock running rightwards, a stationary contact and a left running expansion wave. It is similar to test case 3 but with negative initial velocities. As seen in Fig. 5, the contact is best captured by HLLC scheme and worst by HLL scheme. The AUFS and HLL-CPS-Z_{roe} schemes are also able to capture contact but with little extra diffusion when compared with HLLC scheme. At shock both the AUFS and HLL-CPS-Z_{roe} schemes has little diffusion. The HLL-CPS-Z_{roe} scheme has little extra diffusion at the head of expansion fan. The result of HLL-CPS-Z_{roe} scheme is perfectly monotone; whereas, that of AUFS scheme has a little overshoot at tail of rarefaction wave. The results of AUFS scheme has overshoot at top of fast shock in velocity and pressure plots also.

Test case 6 is of an isolated stationary contact. As seen in Fig. 6, the results of HLL-CPS-Z_{roe}, AUFS and HLLC schemes are coinciding, but that of HLL scheme has considerable amount of diffusion. Thus, the HLL-CPS-Z_{roe} scheme is able to numerically resolve an isolated stationary contact accurately. In fact, the HLL-CPS schemes reduce to exact solution in case of isolated stationary contact.

Test case 7 is of isolated contact moving slowly towards right. As seen in Fig. 7, the results of HLL-CPS-Z_{roe}, HLLC and AUFS schemes have similar amount of diffusion. Whereas, the HLL has lot of diffusion.

Test case 8 is a colliding flow problem. It has two shock waves running outwards either side and a trivial (no density and internal

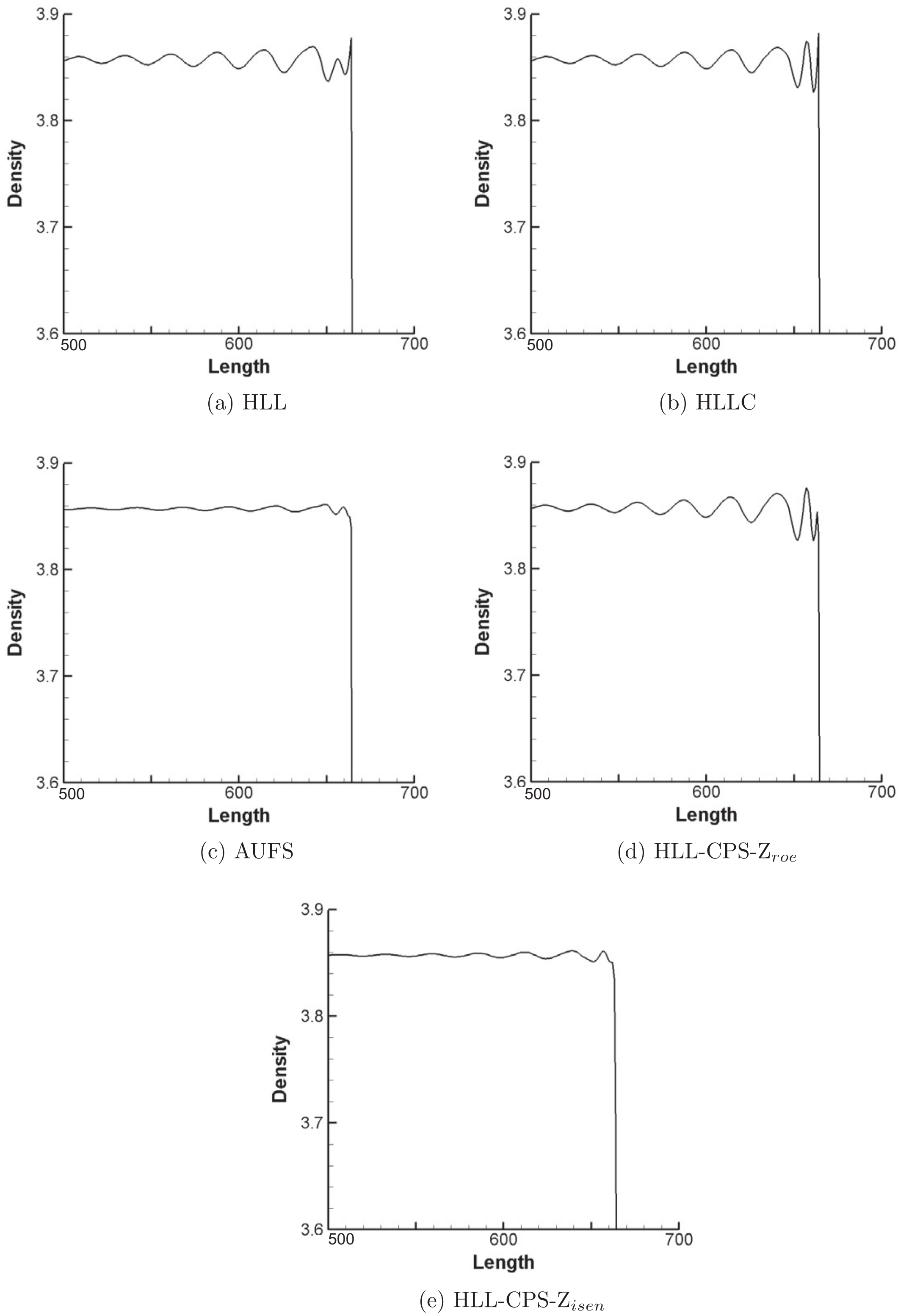


Fig. 9. Fluctuations behind a slowly moving shock.

energy changes across the contact) stationary contact at center. As seen in Fig. 8, density and energy plots show fluctuation at contact for the AUFS, HLLC and HLL-CPS-Z_{roe} schemes. The results of AUFS scheme has fluctuation at both the shocks, which is prominent in density and pressure plots and of lesser magnitude in velocity and internal energy plots. Shock is little smeared in case of AUFS scheme. The HLL scheme resolves the case perfectly.

The proposed scheme has been developed as an alternative to HLLC scheme which can resolve contact accurately. Therefore, the accuracy and efficiency of proposed scheme has been compared with the HLLC scheme. The comparison of operation counts of HLL-CPS-Z_{roe} and HLLC schemes for 1D subsonic flux calculation are given in Appendix B. As seen from the comparison, operation counts of HLL-CPS-Z_{roe} scheme are little more than that of HLLC

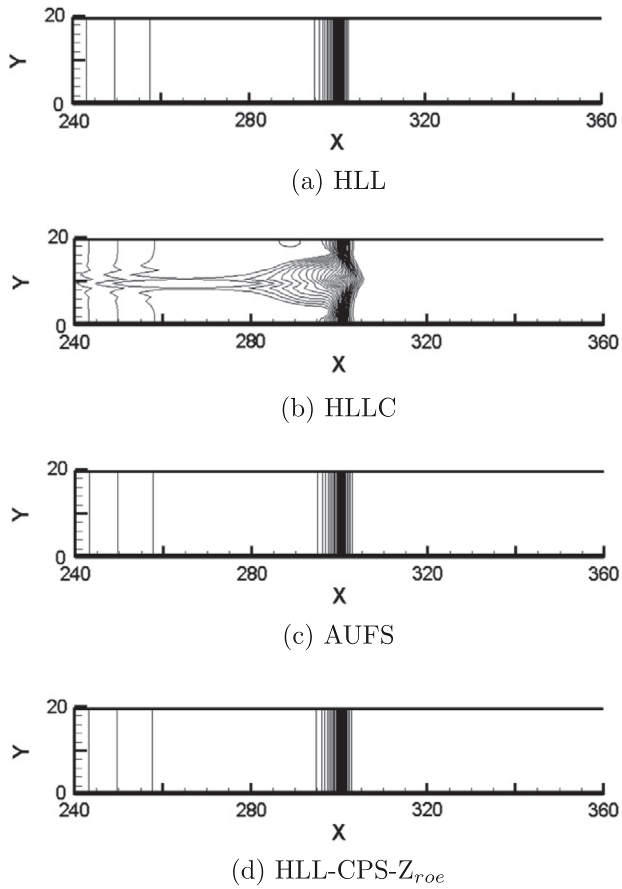


Fig. 10. Density contours for Mach 6 planar shock.

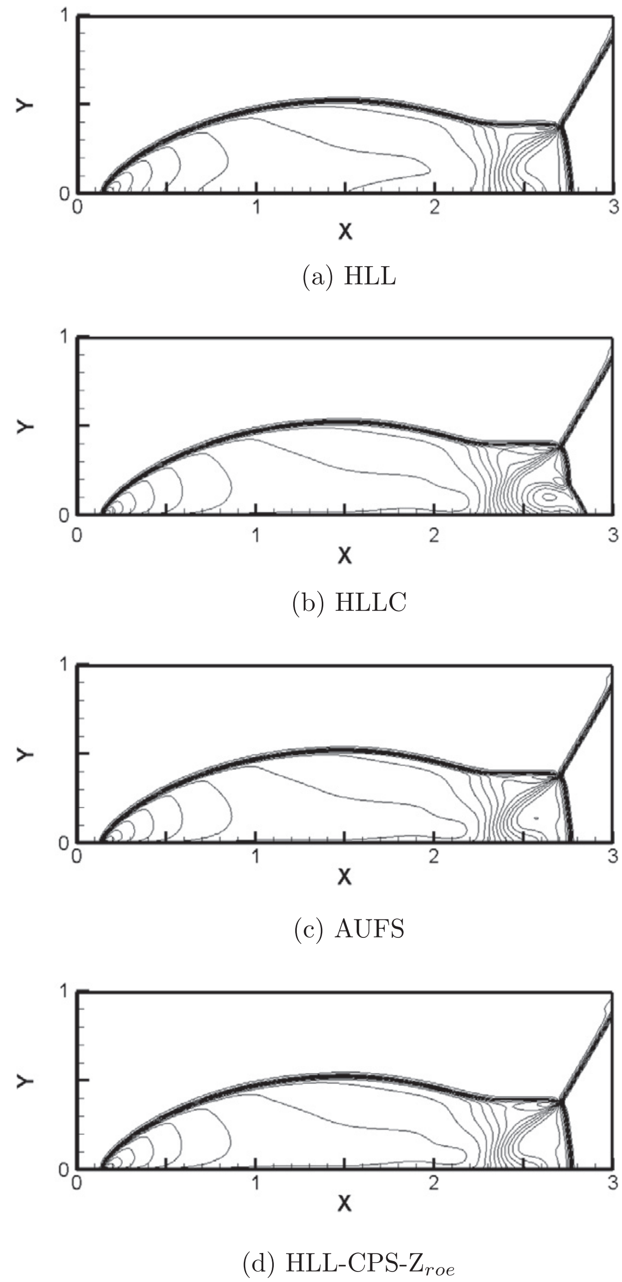


Fig. 11. Density contours for double Mach reflection.

scheme. L_1, L_2, and L_∞ norms have been calculated for above mentioned eight shock tube cases for velocity, density, pressure and specific internal energy for HLLC and HLL-CPS-Z_{roe} schemes. The norms have been calculated based on computations over 2000 cells. The table of norms is given in Appendix C. The norm analysis indicates that, in general, the accuracy of HLL-CPS-Z_{roe} scheme is very similar to that of HLLC.

Considering all eight shock tube cases, it can be summarized that the HLL-CPS-Z_{roe} scheme is able to capture contact accurately. It is slightly more diffusive in shock resolution as compared to HLLC scheme. The HLL-CPS-Z_{roe} scheme is perfectly monotone and does not exhibit any oscillations. The overall accuracy of HLL-CPS-Z_{roe} scheme can be considered to be better than that of HLL and AUFS schemes.

5.1.2. Slowly moving shock problem

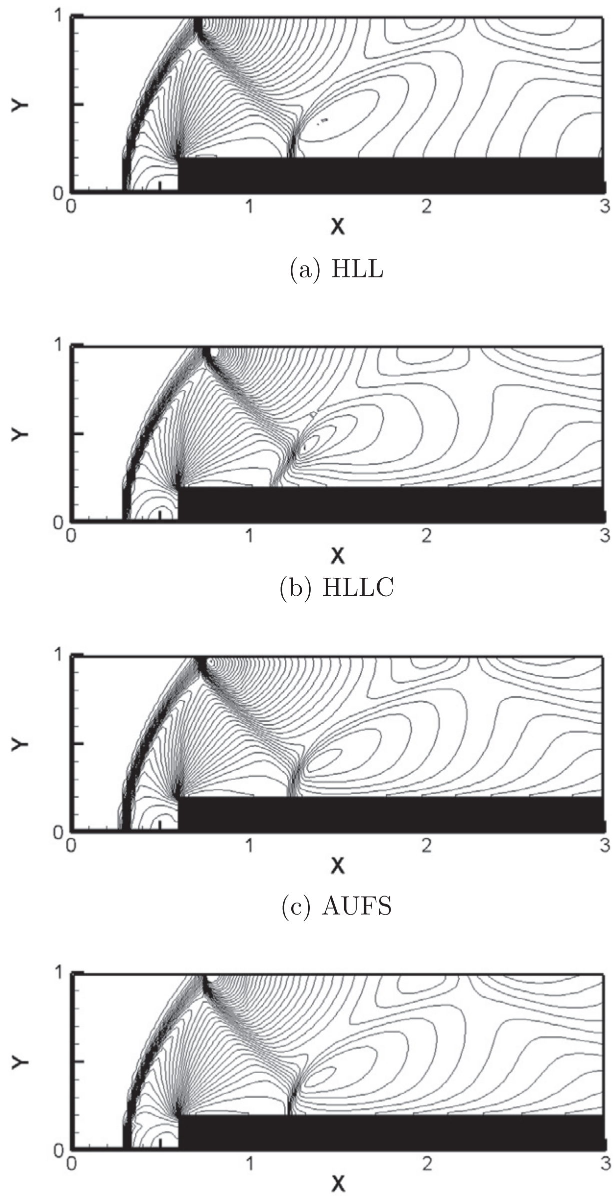
Slowly moving shock problem [24,12] consists of an isolated shock moving very slowly to the right. The domain is one unit long and it has been divided into 1000 cells. The initial position of the shock is at the middle of the domain. The pre shock conditions are $\rho = 1$, $u = -3.44$, $p = 1$ and the post shock conditions are $\rho = 3.86$, $u = -0.81$, $p = 10.33$. The left boundary is set to post shock conditions and right boundary is set to pre shock conditions. The CFL value has been set to unity and solution has been evaluated at $t = 1.5$ units. As seen in Fig. 9, the low frequency post shock fluctuations are of maximum amplitude with HLLC solver. The fluctuations are of reduced amplitude with HLL scheme. The amplitude of fluctuations with HLL-CPS-Z_{roe} solver is more than that of HLL scheme, but less than that of HLLC scheme. In case of AUFS and HLL-CPS-Z_{isen} schemes, these fluctuation are of much reduced amplitude. None of the schemes is truly free from low frequency post shock oscillations.

5.2. Two dimensional test cases

The performance of HLL-CPS-Z_{roe} scheme has been evaluated by solving the problems involving planar shock, double Mach reflection, forward facing step, blunt body and expansion corner. The performance of the HLL-CPS-Z_{roe} scheme is compared with HLL, HLLC and AUFS schemes.

5.2.1. Planar shock problem

The problem consists of a Mach 6 normal shock wave propagating down a rectangular channel [12]. The domain consists of 800×20 cells with unit spacing in both direction. The centerline



(a) HLL

(b) HLLC

(c) AUFS

(d) HLL-CPS-Z_{roe}

Fig. 12. Density contours for Mach 3 flow over a forward facing step.

in y direction (i.e. top edge of tenth cell) has been perturbed to promote odd–even decoupling along the length of shock as

$$y_{i,11} = \begin{cases} y_{i,11} + 0.001 & \text{if } i \text{ is even} \\ y_{i,11} - 0.001 & \text{if } i \text{ is odd} \end{cases}$$

The test case was initially proposed by Quirk [12] and later has been used by many scholars to demonstrate the carbuncle instability. It was reported that those solvers which explicitly identifies the contact would suffer from odd–even decoupling. The domain has been initialized with $\rho = 1.4$, $p = 1$, $u = 0$, and $v = 0$. The inlet boundary condition is set to post shock values, outlet boundary conditions are of zero gradient, top boundary is solid wall and bottom boundary is also a solid wall. The CFL number 0.5 has been used for all the schemes to compute the solutions. The contour plots for density have been shown in Fig. 10 for the time level $t = 55$, where there are 30 contour levels varying from 1.6 to 7.0.

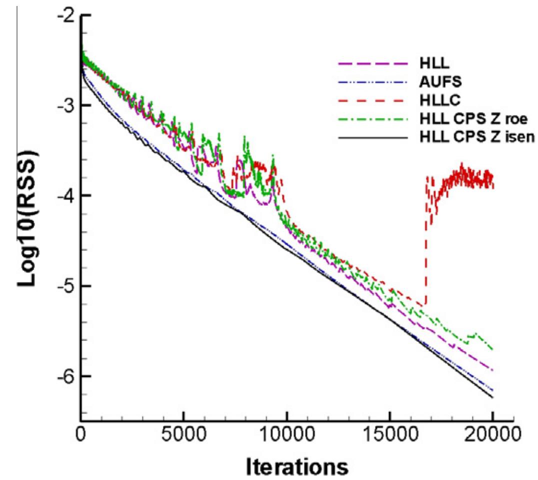


Fig. 13. Convergence history for Mach 20 flow over a blunt body.

Table 2
Comparison of CPU times for blunt body problem.

	HLLC	HLL-CPS-Z_{roe}
CPU time for 15,000 iterations	194.57	195.15
CPU time for residual to fall to 10^{-6}	192.16	182.58
No. of iterations for residual to fall to 10^{-6}	14,814	14,034

Odd–even decoupling is clearly visible with HLLC scheme. The region behind the shock is also polluted. Whereas, HLL, AUFS and the HLL-CPS-Z_{roe} schemes are free from carbuncle instability. The shock front is slightly diffused with AUFS scheme as compared to HLL-CPS-Z_{roe} scheme.

5.2.2. Double Mach reflection problem

Development of kinked Mach stem in double Mach reflection has been reported in various literature [12,25]. Therefore, it is interesting to assess the performance of HLL-CPS scheme for this problem. The problem set up is done by keeping the domain flat and forcing an oblique shock through it [25]. The domain is four units long and one unit wide. The domain has been divided into 480 cells along the length and 120 cells along the width. A Mach 10 shock initially making 60° angle at $x = \frac{1}{6}$ with the bottom reflecting wall is made to propagate through the domain. The cells ahead of shock are initialized to pre shock values ($\rho = 1.4$, $u = 0$, $v = 0$, $p = 1$) and behind the shock are initialized to post shock values. The inlet boundary condition is set to post shock values and outlet boundary condition is set to zero gradient. The top boundary condition is set to simulate the actual shock movement. The bottom boundary up to $x = \frac{1}{6}$ is assigned post shock values and beyond that reflecting wall boundary condition has been applied. The CFL value of 0.8 has been used for all the schemes. The density contours computed by different schemes are shown in Fig. 11, where 30 contour levels varying from 2.0 to 21.5 are used. The plots have been generated for $t = 2.00260 \times 10^{-1}$ units. Though, domain is four units long but plots have been drawn up to three units only.

In Fig. 11, kinked Mach stem is clearly visible in case of HLLC scheme and as a result it has given rise to spurious triple point. The region is polluted in HLLC scheme and the shock is kinked, whereas, HLL, AUFS and HLL-CPS-Z_{roe} schemes have clean performance in the similar region. A small perturbation is seen at top, just ahead of incident oblique shock in case of HLLC, AUFS and HLL-CPS-Z_{roe} schemes. A small spot is seen just ahead of Mach stem

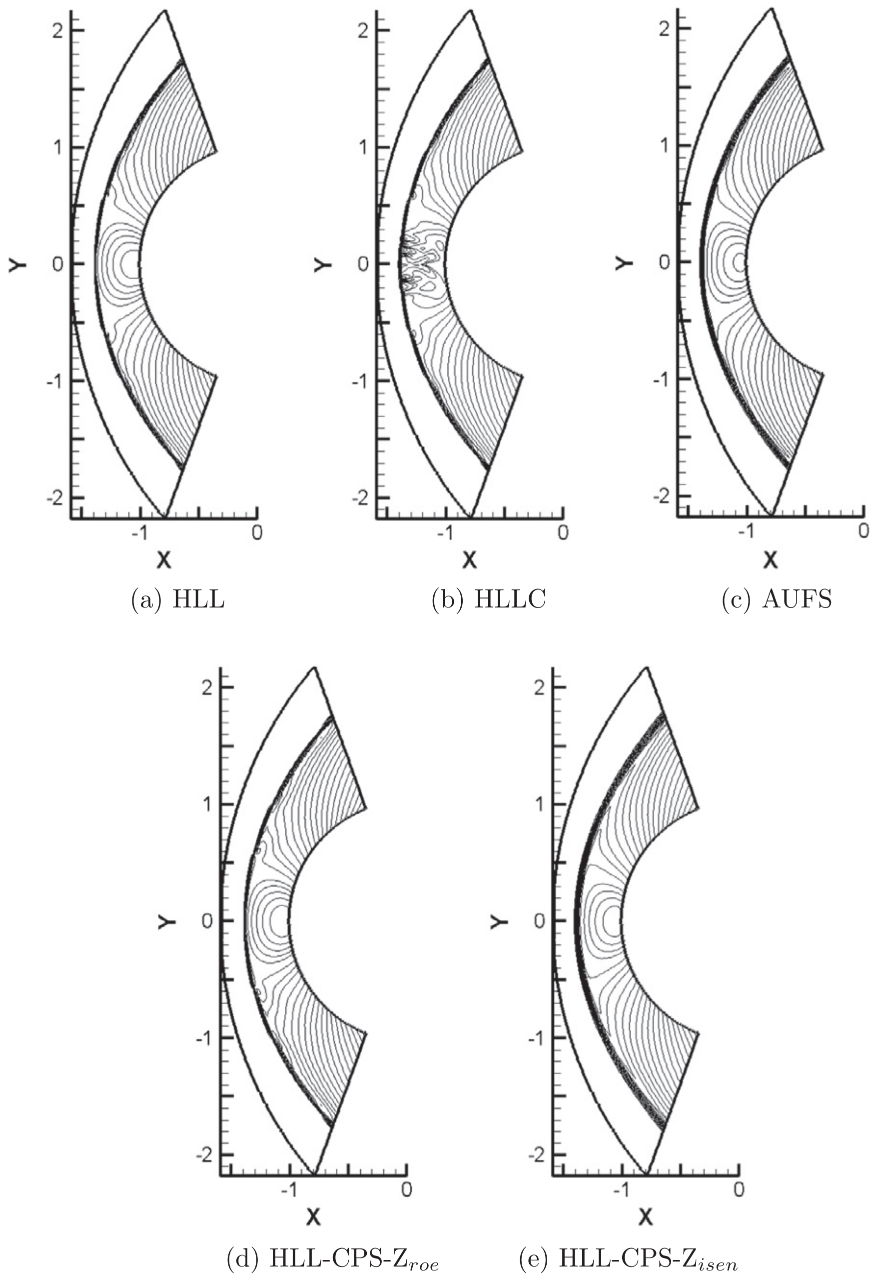


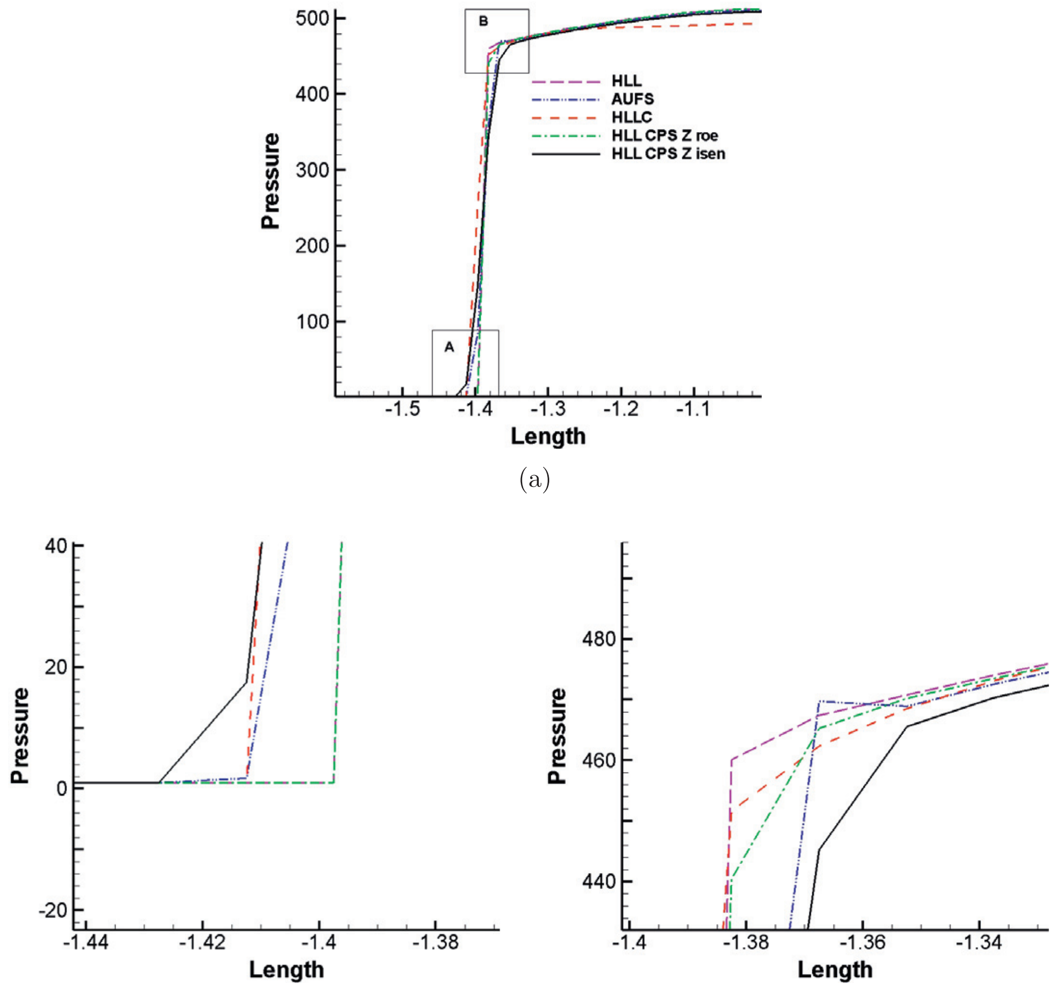
Fig. 14. Density contours for Mach 20 flow over a blunt body.

in case of AUFS scheme, whereas the results of the HLL-CPS-Z$_{roe}$ scheme are as clean as that of HLL scheme.

5.2.3. Forward facing step problem

Mach 3 flow over forward facing step in a 2D channel problem has been studied earlier by Emery [26], Woodward and Colella [25], Batten et al. [27] and many other researchers. The domain is 3 units long and 1 unit wide. A mesh of 120×40 cells as used by Batten et al. is considered. The step is located at 0.6 units downstream of inlet and is 0.2 units high. The complete domain has been initialized with the values $\rho = 1.4$, $p = 1$, $u = 3$, and $v = 0$. The inlet boundary conditions are set equal to freestream conditions, outlet boundary has zero gradient, bottom and top boundary have been given reflecting solid wall boundary conditions. The CFL value has been taken as 0.5. Fig. 12 shows the plots generated at time $t = 4.0$ units. A total of 45 density contour levels varying from 0.2 to 7.0 have been illustrated.

In the Fig. 12, it is observed that all the schemes are able to capture primary bow and reflected shocks. The bow shock wave captured by HLLC, HLL and HLL-CPS-Z$_{roe}$ schemes is crisp whereas it is slightly diffused in case of AUFS scheme especially at the foot of the bow shock. All schemes are able to resolve Mach stem marginally. The first reflected shock is diffused in case of HLL scheme as compared to other schemes. The second reflected shock is crisp in case of HLLC, HLL and HLL-CPS-Z$_{roe}$ schemes, whereas it is little diffused in case of AUFS scheme. In case of HLLC scheme it is badly kinked, whereas the HLL-CPS-Z$_{roe}$ scheme does not show any sign of instability.



(a)

(b) Zoomed portion A

(c) Zoomed portion B

Fig. 15. Pressure variation along the line of symmetry for flow over blunt body.

5.2.4. Blunt body problem

Hypersonic flow over a blunt body is a typical problem to assess the performance of a scheme with respect to carbuncle instability [12,8,13]. Mach 20 flow has been simulated over the blunt body. The grid for the blunt body has been generated using an algorithm given in reference [8].

The computational domain consists of 40×320 cells. The domain has been initialized with values $\rho = 1.4$, $p = 1$, $u = 20$, and $v = 0$. The inlet boundary conditions have been set to freestream values. The flow being supersonic, top and bottom boundary values are extrapolated. The solid wall condition has been applied on the blunt body. Though the problem is steady, the residue of the HLLC scheme do not drop below three decades even after 35,000 iterations and keeps oscillating. Whereas residue for all other schemes converged to 1×10^{-6} close to 20,000 iterations. Therefore, output after 20,000 iterations has been considered for comparison of all the schemes. The convergence histories, based on L2-norm of density error, for all the schemes are illustrated in Fig. 13.

CPU time is a parameter used to assess the efficiency of a scheme. The CPU times for this (steady state) problem have been compared in the Table 2. The computations have been done on a laptop with Intel (R) Core (TM) i3 CPU M 330 @ 2.13 GHz with 6 GB DDR3 SDRAM on Ubuntu 11.10 operating system.

Based on the operation counts (refer to Appendix B) and the CPU times, it can be summarized that although the operation counts of HLL-CPS-Z_{roe} scheme are little more than that of HLLC scheme, HLL-CPS-Z_{roe} scheme is more efficient due to faster convergence rate (as seen in Fig. 13).

The contour plots for density obtained from various schemes are shown in Fig. 14, where a total of 27 density contours varying from 2 to 8.7 have been drawn. As seen in Fig. 14, the result of HLL-CPS-Z_{roe} scheme is found to be almost identical to that of HLL scheme. The HLLC scheme shows carbuncle phenomenon at the center. The HLL, AUFS, HLL-CPS-Z_{roe} and HLL-CPS-Z_{isen} schemes are free from carbuncle. The HLL, HLLC and HLL-CPS-Z_{roe} schemes have been able to resolve the bow shock crisply, whereas the same is diffused in case of AUFS and HLL-CPS-Z_{isen} schemes. The HLL and HLL-CPS-Z_{roe} schemes have a very little waviness in the sonic region close to the bow shock, whereas the same is absent in AUFS and HLL-CPS-Z_{isen} schemes.

The pressure plot along the line of symmetry of blunt body is given in Fig. 15. The result of HLLC scheme is found to be poor due to carbuncle. The result of HLL and HLL-CPS-Z_{roe} schemes are very crisp and exactly coinciding at the bottom of the curve (see Fig. 15b), whereas the results of AUFS and HLL-CPS-Z_{isen} schemes are little diffusive. At the top of the curve (see Fig. 15c), the HLL-CPS-Z_{roe} scheme is very slightly diffusive as compared to the HLL scheme. The AUFS and HLL-CPS-Z_{isen} schemes are more diffusive as seen in Fig. 15c. The AUFS scheme has a kink as seen in Fig. 15c, whereas HLL-CPS-Z_{roe} scheme is perfectly monotone with good accuracy.

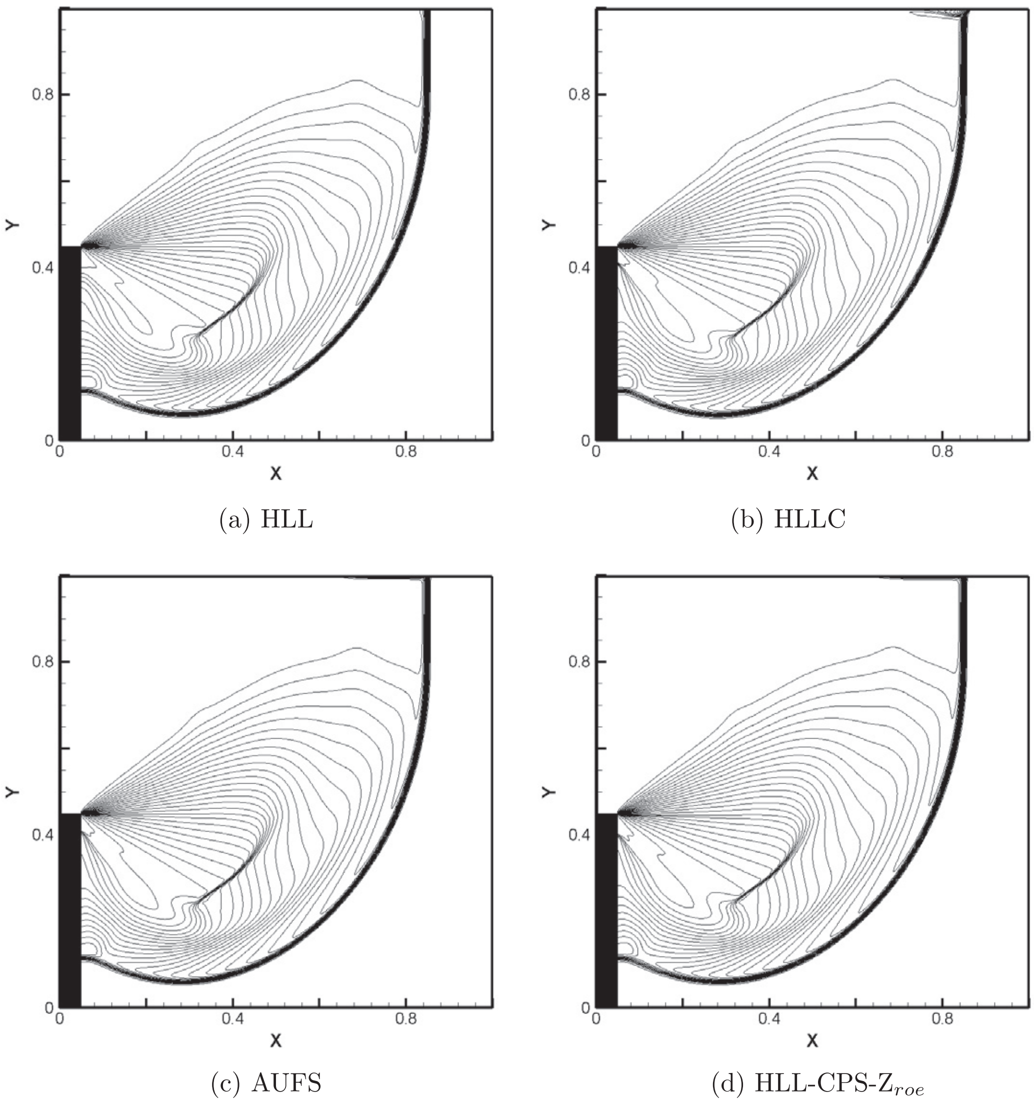


Fig. 16. Density contours for diffraction of Mach 5.09 normal shock around a 90° corner.

Table 3
Comparison of operation counts for 1D subsonic flux calculation.

	HLLC	HLL-CPS-Z_{roe}
Addition/Subtraction	35	35
Multiplication/Division	51	66
Square root	8	8
Comparison	6	6
Assignment	32	39

Table 4
Norm results for shock tube test cases 1 and 2.

		HLLC			HLL-CPS-Z_{roe}		
		L_1	L_2	L_∞	L_1	L_2	L_∞
Case 1	ρ	0.002269	0.000255	0.140112	0.002412	0.000257	0.146491
	u	0.001395	0.000537	1.020177	0.001553	0.000531	1.016944
	p	0.000778	0.000126	0.227101	0.000961	0.000135	0.235757
	e	0.011742	0.001662	1.084559	0.011909	0.001660	1.091794
Case 2	ρ	0.002425	0.000100	0.028145	0.002362	0.000098	0.028085
	u	0.009606	0.000360	0.071073	0.008980	0.000336	0.067187
	p	0.001113	0.000049	0.015668	0.001095	0.000049	0.015635
	e	0.023903	0.001384	0.471950	0.021994	0.001226	0.425613

5.2.5. Supersonic corner problem

Supersonic corner problem was first used by Quirk [12] and later by many other scholars [8] to test the performance of numerical schemes. This problem consists of sudden expansion of Mach 5.09 normal shock around a 90° corner. The domain is a square of one unit which is divided into 400 × 400 cells. The corner is located at $x = 0.05$ and $y = 0.45$. The initial location of the normal shock is at $x = 0.05$. The domain to the right of the shock is assigned initially with pre shock conditions of $\rho = 1.4$, $p = 1$, $u = 0$, and $v = 0$ and domain to the left of the shock with calculated post shock conditions. The inlet boundary is supersonic, outlet boundary has zero gradient and bottom boundary behind the corner uses extrapolated values. The corner has reflective wall boundary condition. The top boundary simulates the interface of boundary with the shock which is determined based on shock speed. The CFL value of 0.8 is considered here. The plots shown in Fig. 16 have been generated at time $t = 0.1561$ units. A total of 30 density contour levels varying from 0 to 7.1 have been illustrated.

The solution of HLLC scheme shows instability ahead of shock at the top. The result of HLL scheme is perfectly clean, whereas a little benign oscillation is seen at the top just ahead of shock in case of

Table 5
Norm results for shock tube test cases 3–6.

		HLLC			HLL-CPS-Z_{roe}		
		L_1	L_2	L_∞	L_1	L_2	L_∞
Case 3	ρ	0.044360	0.006219	4.192880	0.044979	0.006333	4.345100
	u	0.064722	0.013566	18.07942	0.068851	0.014390	18.38919
	p	1.498784	0.250102	374.7849	1.546533	0.270225	390.9462
	e	22.03850	3.407337	1623.959	21.91064	3.382862	1620.196
Case 4	ρ	0.155369	0.019861	11.03497	0.154715	0.019771	10.93135
	u	0.010428	0.005757	9.489574	0.009743	0.005438	8.840403
	p	1.153931	0.561090	841.0129	1.125776	0.538581	841.1079
	e	1.541290	0.208350	105.6690	1.534733	0.207664	105.6425
Case 5	ρ	0.002928	0.001303	2.245259	0.004238	0.001713	2.543548
	u	0.038540	0.007165	13.90339	0.046967	0.007212	13.81075
	p	0.958448	0.113982	206.4247	1.199668	0.122665	214.6543
	e	1.487909	0.251996	462.1417	2.296461	0.626842	1201.169
Case 6	ρ	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000
	u	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000
	p	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000
	e	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000

Table 6
Norm results for shock tube test cases 7 and 8.

		HLLC			HLL-CPS-Z_{roe}		
		L_1	L_2	L_∞	L_1	L_2	L_∞
Case 7	ρ	0.003077	0.000424	0.196574	0.003077	0.000424	0.196574
	u	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000
	p	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000
	e	0.005606	0.000784	0.408219	0.005606	0.000784	0.408219
Case 8	ρ	0.007137	0.001090	1.467372	0.002621	0.001052	1.434542
	u	0.001290	0.000867	1.226812	0.001178	0.000787	1.113436
	p	0.002236	0.001236	1.704433	0.002112	0.001073	1.463592
	e	0.003931	0.001123	1.578294	0.001742	0.000961	1.355633

AUFS and HLL-CPS-Z_{roe} schemes. The benign oscillations subside for higher order results in case of AUFS and HLL-CPS-Z_{roe} schemes, but it aggravates in case of HLLC scheme. The HLL-CPS-Z_{roe} scheme is found to be free from the problem of expansion shock and negative internal energy.

6. Conclusion

A novel Riemann solver for Euler equations, called HLL-CPS scheme, is developed which can resolve contact discontinuity very accurately. As the HLL framework is used, the present method turned out to be very simple and economical. Since the contact wave is neglected when the HLL solver is formulated for Euler equations in its original conservative form, the Euler equations are split into convective and pressure parts (similar to Zha Bilgen, AUSM and CUSP schemes) before formulating the HLL-type scheme in order to recover the contact wave in the present method. The ability to resolve the contact is further enhanced by modifying the dissipation term of HLL flux for the pressure split term. The present method is able to successfully solve tough 1D test problems proposed by Toro [23] and Quirk [24], and 2D test problems proposed by Quirk [12] and Emery [26], which are too severe for many advanced solvers to survive. The present method is found to be free from the problems of expansion shock, negative internal energy, carbuncle phenomenon, kinked Mach stem, odd even decoupling. The sonic point, contact and expansion fan are resolved by the present method with the accuracy similar to AUFS scheme; whereas shocks resolved by the present method are better than AUFS scheme. In case of 1D test cases 3, 5, and 8 involving shock tube, the HLL-CPS results are found to be perfectly monotone, whereas the AUFS results show oscillations close to expansion fans. In case of 1D receding flow problem, the HLL-CPS result is slightly inferior as compared to the AUFS result. This advantage of AUFS scheme is limited to first order only. In case of slowly moving shock, none of the methods is able to produce perfectly monotone solution; the AUFS method produces best solution followed by HLL-CPS-Z_{isen} scheme. For blunt body problem, the results of HLL-CPS-Z_{roe} and HLL schemes are found to be more monotone and accurate, while the solution of the AUFS scheme shows a slight kink in the pressure plot along the line of symmetry. Thus, the proposed HLL-CPS method is found to be quite robust and accurate.

Since a contact discontinuity is considered to be numerically equivalent to a limiting case of a viscous boundary layer [28], the accuracy of a numerical method for Navier–Stokes equations can be indirectly verified by the contact discontinuity problem. Therefore, the present scheme is expected to produce accurate results for viscous computations. The next logical step would therefore be to extend the present scheme for Navier–Stokes equations.

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Appendix A. Algorithm for 1D HLL-CPS-Z_{roe}

γ = Ratio of specific heats.

<u>Calculation for left state</u>

ρ_L = density of ith cell
u_L = velocity of ith cell
e_L = internal energy of ith cell
$p_L = (\gamma - 1) * (e_L - 0.5 * \rho_L * u_L * u_L)$
$h_L = (e_L + p_L)/\rho_L$
$a_L = \sqrt{\gamma * p_L/\rho_L}$

<u>Calculation for right state</u>

ρ_R = density of $(i+1)$th cell
u_R = velocity of $(i+1)$th cell
e_R = internal energy of $(i+1)$th cell
$p_R = (\gamma - 1) * (e_R - 0.5 * \rho_R * u_R * u_R)$
$h_R = (e_R + p_R)/\rho_R$
$a_R = \sqrt{\gamma * p_R/\rho_R}$

<u>Allied calculations</u>

$r = \sqrt{\rho_R/\rho_R}$
$\tilde{u} = (u_L + r * u_R)/(1 + r)$
$\tilde{h} = (h_L + r * h_R)/(1 + r)$
$\tilde{a} = \sqrt{(\gamma - 1) * (\tilde{h} - 0.5 * \tilde{u} * \tilde{u})}$
$\bar{u} = (u_L + u_R)/2$
$\bar{a} = (a_L + a_R)/2$

<u>Calculation of wavespeeds</u>

$S_L = min(0, \quad u_L - a_L, \quad \tilde{u} - \tilde{a})$
$S_R = max(0, \quad u_R + a_R, \quad \tilde{u} + \tilde{a})$
$if(\bar{u} = 0)\{S_L = -\bar{a}, \quad S_R = \bar{a}\}$

<u>Calculation of flux 1</u>

$if(\bar{u} \geqslant 0)$
{
$M = \bar{u}/(\bar{u} - S_L)$
$flux1[0] = M*(\rho_L) * (u_L - S_L)$
$flux1[1] = M * (\rho_L * u_L) * (u_L - S_L)$

$flux1[2] = M * (e_L) * (u_L - S_L)$
}
else
{
$M = \bar{u}/(\bar{u} - S_R)$
$flux1[0] = M * (\rho_R) * (u_R - S_R)$
$flux1[1] = M * (\rho_R * u_R) * (u_R - S_R)$
$flux1[2] = M * (e_R) * (u_R - S_R)$
}

Calculation of flux 2
$flux2[0] = [(S_R * S_L)/(S_R - S_L)] * [(p_L - p_R)/(\bar{a} * \bar{a})]$
$flux2[1] = 0.5 * (p_L + p_R) + 0.5 * [(S_R + S_L)/(S_R - S_L)] * (p_L - p_R) - [(S_R * S_L)/(S_R - S_L)] * [(p_L * u_L - p_R * u_R)/(\bar{a} * \bar{a})]$
$flux2[2] = 0.5 * (p_L * u_L + p_R * u_R) + 0.5 * [(S_R + S_L)/(S_R - S_L)] * (p_L * u_L - p_R * u_R) - [(S_R * S_L)/(S_R - S_L)] * [\{(\bar{a} * \bar{a})/(\gamma - 1)\} * (p_L - p_R) + 0.5 * (p_L * u_L * u_L - p_R * u_R * u_R)]/(\bar{a} * \bar{a})$

Calculation of total flux
$flux[0] = flux1[0] + flux2[0]$
$flux[1] = flux1[1] + flux2[1]$
$flux[2] = flux1[2] + flux2[2]$

Appendix B. Operation counts for 1D subsonic flux calculation

See Table 3.

Appendix C. Norm calculation for 1D shock tube cases

The formulae used for norm calculation are

$$L_1 = \frac{\sum_{i=1}^{N}[abs(X_{i_{numerical}} - X_{i_{exact}})]}{N}$$
$$L_2 = \frac{\sqrt{\sum_{i=1}^{N}[X_{i_{numerical}} - X_{i_{exact}}]^2}}{N}$$
$$L_\infty = max[abs(X_{i_{numerical}} - X_{i_{exact}})]$$

where subscript *numerical* indicates the numerically calculated value of a variable and subscript *exact* indicates the variable value calculated using exact solution. Number of cells (N) has been taken as 2000 (Tables 4–6).

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